ME2201: Kinematics and Dynamics of Machinery

Tutorial 3

Submission Date: 19-Aug-2026, in class

1. Consider the six-link mechanism shown in Figure 1. If the crank 2 is rotating at an angular velocity of 5 rad/s at the instant shown, what is the velocity of the slider?

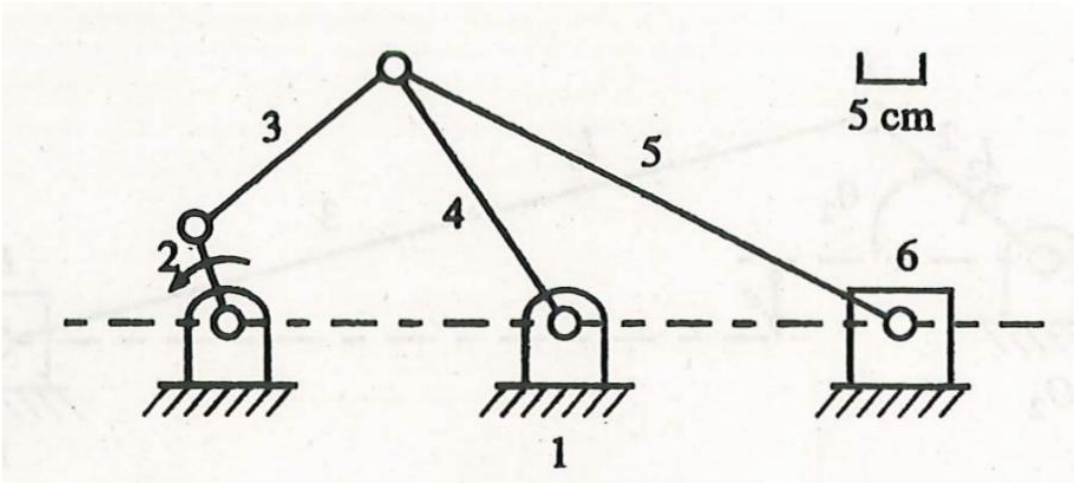


Figure 1: Six-link mechanism

2. Consider the kinematic diagram of the mechanism shown in Figure 2 (not drawn to scale). Determine the mobility of the mechanism. If link 2 is rotating at an angular velocity of 20 rpm in the counter-clockwise direction, what are the velocities of sliders 4 and 6? Link lengths are as follows: O_2A = 12 cm, AB = 48 cm, BC = 18 cm, and CD = 18 cm.

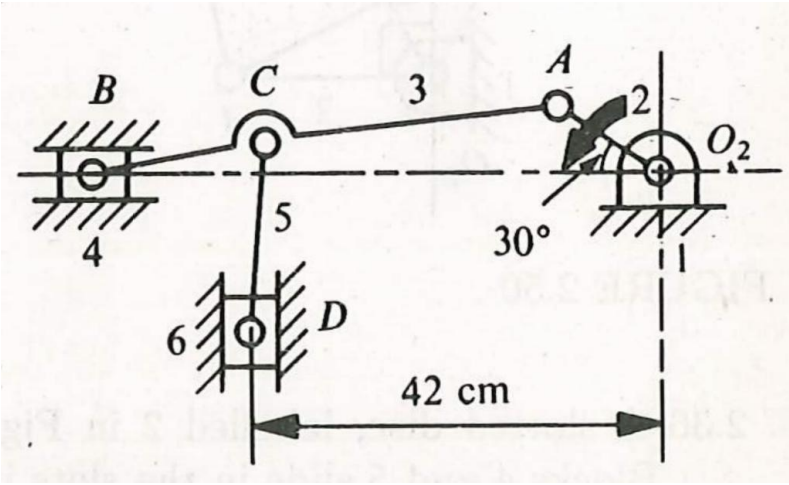


Figure 2

3. (Optional) The kinematic diagram of a mechanism is shown in Figure 3 (diagram not to scale). Link 5 rotates at 1 rad/s, and the hydraulic cylinder-piston system OC_7 extends at 10 cm/s. Given O_2A = 10 cm, AB = 12 cm, O_7C = 14 cm, and O_5D = 15 cm. For the velocity diagram, choose a scale of 1 cm = 5 cm/s. Using the graphical method, answer the following questions.

a. Find the angular velocity of link 7 at this instant.
b. Find the angular velocity of link 2 at this instant.

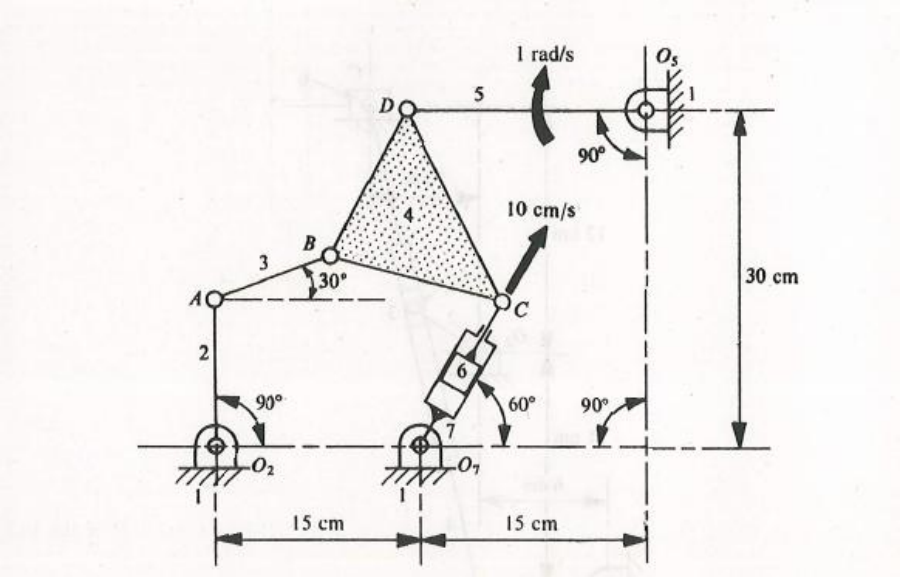


Figure 3